
Your Full Project Title Here

Author One Author Two Author Three

Abstract

This is a basic LaTeX template for CSCI 567 project report. It is based on the ICML 2021 LaTeX template.

1. Style Guide

Below are examples of how to format sections, add citations, figures, and tables. All margins, font sizes, etc. are already pre-specified in the style file, which you should not change at all. You should only edit the main.tex file by replacing the author, title, and text of the paper with your own.

The formatting instructions contained in the style file are summarized in the sections below. You do not need to read these in detail, but they can be a helpful example of L^AT_EX formatting for your reference.

2. Partitioning the Text

You should organize your paper into sections and paragraphs to help readers place a structure on the material and understand its contributions.

2.1. Sections and Subsections

Section headings should be numbered, flush left, and set in 11 pt bold type with the content words capitalized. Leave 0.25 inches of space before the heading and 0.15 inches after the heading.

Similarly, subsection headings should be numbered, flush left, and set in 10 pt bold type with the content words capitalized. Leave 0.2 inches of space before the heading and 0.13 inches afterward.

Finally, subsubsection headings should be numbered, flush left, and set in 10 pt small caps with the content words capitalized. Leave 0.18 inches of space before the heading and 0.1 inches after the heading.

Please use no more than three levels of headings.

2.2. Paragraphs and Footnotes

Within each section or subsection, you should further partition the paper into paragraphs. Do not indent the first

line of a given paragraph, but insert a blank line between succeeding ones.

You can use footnotes¹ to provide readers with additional information about a topic without interrupting the flow of the paper. Indicate footnotes with a number in the text where the point is most relevant. Place the footnote in 9 point type at the bottom of the column in which it appears. Precede the first footnote in a column with a horizontal rule of 0.8 inches.²

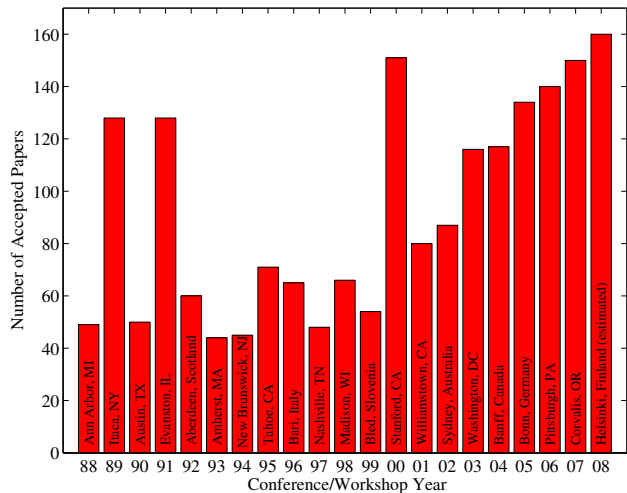


Figure 1. Historical locations and number of accepted papers for International Machine Learning Conferences (ICML 1993 – ICML 2008) and International Workshops on Machine Learning (ML 1988 – ML 1992). At the time this figure was produced, the number of accepted papers for ICML 2008 was unknown and instead estimated.

3. Figures

You may want to include figures in the paper to illustrate your approach and results. Such artwork should be centered, legible, and separated from the text. Lines should be dark and at least 0.5 points thick for purposes of reproduction,

¹Footnotes should be complete sentences.

²Multiple footnotes can appear in each column, in the same order as they appear in the text, but spread them across columns and pages if possible.

Algorithm 1 Bubble Sort

Input: data x_i , size m **repeat**Initialize $noChange = true$.**for** $i = 1$ **to** $m - 1$ **do****if** $x_i > x_{i+1}$ **then**Swap x_i and x_{i+1} $noChange = false$ **end if****end for****until** $noChange$ is $true$

Table 1. Classification accuracies for naive Bayes and flexible Bayes on various data sets.

DATA SET	NAIVE	FLEXIBLE	BETTER?
BREAST	95.9 $\pm$ 0.2	96.7 $\pm$ 0.2	✓
CLEVELAND	83.3 $\pm$ 0.6	80.0 $\pm$ 0.6	×
GLASS2	61.9 $\pm$ 1.4	83.8 $\pm$ 0.7	✓
CREDIT	74.8 $\pm$ 0.5	78.3 $\pm$ 0.6	
HORSE	73.3 $\pm$ 0.9	69.7 $\pm$ 1.0	×
META	67.1 $\pm$ 0.6	76.5 $\pm$ 0.5	✓
PIMA	75.1 $\pm$ 0.6	73.9 $\pm$ 0.5	
VEHICLE	44.9 $\pm$ 0.6	61.5 $\pm$ 0.4	✓

and text should not appear on a gray background.

Label all distinct components of each figure. If the figure takes the form of a graph, then give a name for each axis and include a legend that briefly describes each curve. Do not include a title inside the figure; instead, the caption should serve this function.

Number figures sequentially, placing the figure number and caption *after* the graphics, with at least 0.1 inches of space before the caption and 0.1 inches after it, as in Figure 1. The figure caption should be set in 9 point type and centered unless it runs two or more lines, in which case it should be flush left. You may float figures to the top or bottom of a column, and you may set wide figures across both columns (use the environment `figure*` in L^AT_EX). Always place two-column figures at the top or bottom of the page.

4. Algorithms

If you are using L^AT_EX, please use the “algorithm” and “algorithmic” environments to format pseudocode. These require the corresponding stylefiles, `algorithm.sty` and `algorithmic.sty`, which are supplied with this package. Algorithm 1 shows an example.

5. Tables

You may also want to include tables that summarize material. Like figures, these should be centered, legible, and numbered consecutively. However, place the title *above* the table with at least 0.1 inches of space before the title and the same after it, as in Table 1. The table title should be set in 9 point type and centered unless it runs two or more lines, in which case it should be flush left.

Tables contain textual material, whereas figures contain graphical material. Specify the contents of each row and column in the table’s topmost row. Again, you may float tables to a column’s top or bottom, and set wide tables across both columns. Place two-column tables at the top or bottom of the page.

6. Citations and References

Please use APA reference format regardless of your formatter or word processor. If you rely on the L^AT_EX bibliographic facility, use `natbib.sty` and `icml2021.bst` included in the style-file package to obtain this format.

Citations within the text should include the authors’ last names and year. If the authors’ names are included in the sentence, place only the year in parentheses, for example when referencing Arthur Samuel’s pioneering work (1959). Otherwise place the entire reference in parentheses with the authors and year separated by a comma (Samuel, 1959). List multiple references separated by semicolons (Kearns, 1989; Samuel, 1959; Mitchell, 1980). Use the ‘et al.’ construct only for citations with three or more authors or after listing all authors to a publication in an earlier reference (Michalski et al., 1983).

7. Final instructions

Do not change any aspects of the formatting parameters in the style files. In particular, do not modify the width or length of the rectangle the text should fit into, and do not change font sizes (except perhaps in the **References** section; see below). Please note that pages should be numbered.

7.1. Preparing PDF files

You can download a PDF version of your report to submit to Gradescope with the download PDF icon above the PDF preview in Overleaf.

References

- Kearns, M. J. *Computational Complexity of Machine Learning*. PhD thesis, Department of Computer Science, Harvard University, 1989.
- Michalski, R. S., Carbonell, J. G., and Mitchell, T. M. (eds.). *Machine Learning: An Artificial Intelligence Approach, Vol. I*. Tioga, Palo Alto, CA, 1983.

Mitchell, T. M. The need for biases in learning generalizations. Technical report, Computer Science Department, Rutgers University, New Brunswick, MA, 1980.

Samuel, A. L. Some studies in machine learning using the game of checkers. *IBM Journal of Research and Development*, 3(3):211–229, 1959.